
TECHNICAL SKILLS

Languages : C++, Python, MATLAB, Dart, SQL

Robotics & Simulation : ROS2, RViz, Gazebo, MuJoCo, Simulink, Simscape

ML & CV Libraries : NumPy, PyTorch, Pandas, Keras, Tensorflow, OpenCV, Matplotlib, Seaborn, Scikit-Learn

Frameworks & Tools: Flutter, Github, Gitlab, Firebase

PROFESSIONAL EXPERIENCE

Software & Automation Developer

India

Rashmi Enterprises | Jan 2021 - July 2023

Technologies: ROS 2, Gazebo, OpenCV, Python, AMR Simulation, Git, CI/CD

- Led end-to-end software initiatives—from factory automation (ROS 2) to customer-facing **web platform** (+50 % inquiries).
- Prototyped an **OpenCV print-defect detector** (≈95 % accuracy at 20 FPS) for real-time quality control.
- Simulated & tuned a **5-AMR fleet in ROS 2 + Gazebo**, enabling autonomous material flow across production lines.
- Deployed a **vision-guided sorting cell** (>90% classification accuracy) with ROS 2-controlled actuators.
- Modularized the **ROS 2 stack (perception/ planning/ control)**, allowing multi-station testing and reuse.

PROJECTS

HELIOS - Autonomous SAR Drone (Ongoing)

[Github](#) | [Demo](#)

ROS2, C++, Python, Gazebo, Linux

- **Reduced SLAM drift to <5%**, measured by loop-closure error, by **fusing LiDAR, camera, and IMU** in ROS2 pipelines.
- Achieved **~90% RRT* success** in ≤2.5s by **tuning planner heuristics and optimizing collision checks**.
- Enabled **stable flight** with a **custom C++ ROS 2-Gazebo plugin** for **PID-tuned thrust/velocity control**.
- Reached **~80% victim detection accuracy** by combining **OpenCV-based vision** with **mission-aware path replanning**.

ADAS Stack – Lane Keeping & Emergency Braking

[Github](#) | [Demo](#)

ROS2, Python, Carla, Linux

- Held **<25 cm lane error at 60 km/h** using **pure-pursuit** with **adaptive lookahead**.
- Triggered **AEB in <500 ms**, measured by TTC, using **YOLOv8n** with class-aware tracking.
- Applied **progressive slowing**, measured by **obstacle distance bands**, by adjusting **PID targets** based on **proximity**.
- Developed a **real-time lane detection pipeline (HSV→ROI→polyfit)** with **temporal smoothing**.
- Validated system in **40+ mixed-traffic CARLA scenarios** with **modular ROS 2 nodes for perception and control**.

Real Steel – Vision-to-Motion Robot Tracker

[Github](#) | [Demo](#)

ROS2, Python, MuJoCo, Linux

- Achieved **<100 ms latency** and **20 Hz control loop** by mapping human keypoints to **7-DOF robot joints** using MediaPipe.
- Delivered **~90% smoothness in motion transitions** by engineering a **constraint-aware IK solver** with joint-limit handling.
- Enabled **real-time simulation of responsive robot sparring** by integrating **vision, IK, and joint-space control modules**.

GENESIS – Real-World Sensor Enhancement

[Github](#) | [Demo](#)

Python, PyTorch

- Achieved **0.89 STOI (Short Time Objective Intelligibility)** and **1.92 dB SSNR (Spectral Signal-to-Noise Ratio)** by applying a **CNN-Conformer GAN** to denoise real-world audio signals.
- Demonstrated **ML-based signal enhancement** on 11k+ samples, simulating **edge-case sensor degradation in noisy conditions**.

EDUCATION

Iowa State University

MS in Computer Science (CGPA: 3.72/4)

Ames, IA

Aug 2023 - Aug 2025